

System Identification and Development-Stage NMPC for a CloudPendulum Double Pendulum

Abhishek Alaya Udupa, Sichen Li, Venkateswarlu Reddy Konkala, and Xiaojie Ning

Abstract—Swing-up of a remote, underactuated double pendulum is sensitive to model mismatch, while the hardware parameters are unknown. We identify a base-parameter model of a CloudPendulum cell from real free-swing, D-optimal, and high-speed chirp data, and validate it by single-step and finite-horizon prediction error. An acados SQP-RTI nonlinear model predictive controller (NMPC) tracks an offline-optimized reference at a target rate of 400 Hz. During controller development, a bounded 5 mNm assist on the passive joint enabled acrobot and pendubot swing-up, upright holding, and recovery in scheduled disturbance tests. These runs are not evaluation-compliant because passive-joint friction compensation is prohibited in the competition evaluation. Strictly uncompensated pendubot trials departed reproducibly from the reference near a constraint-active segment, motivating further compliant-controller development.

Index Terms—system identification, underactuated robots, nonlinear model predictive control, acrobot, pendubot, remote hardware

I. INTRODUCTION

The CloudPendulum platform exposes physical double-pendulum cells over a network API and supports the RealAIGym and AI Olympics benchmarking workflow [1]–[4]. The task is classical: swing the pendulum from hanging to upright and keep it there with a single actuated joint [5], [6]. Four properties make it hard in practice: the plant parameters (especially friction) are unknown and cell-specific; only one joint is actuated, so the passive joint is reached only through coupling; the hardware enforces measured torque and velocity limits under network latency; and remote commissioning time is scarce.

This report contributes: (i) a base-parameter identification pipeline driven entirely by real hardware data, with identifiability analysis; (ii) single-step and MPC-horizon model validation; (iii) development-stage NMPC experiments on both configurations with a bounded 5 mNm passive-joint assist; and (iv) a diagnosis of strictly uncompensated pendubot failures. The assisted runs isolate the influence of passive-joint friction but are not valid qualification results under the single-actuator evaluation rule.

II. SYSTEM DESCRIPTION AND CONSTRAINTS

The plant state is $x = [q_1, q_2, \dot{q}_1, \dot{q}_2]^\top$ (shoulder angle from hanging, relative elbow angle) with input $u = [\tau_1, \tau_2]^\top$

The authors are with Chalmers University of Technology, Gothenburg, Sweden. Emails: alayaa@chalmers.se, sichenli@chalmers.se, konkala@chalmers.se, and xiaojie@chalmers.se.

This report was prepared for TRA455 Artistic Intelligence in Robotics. All experiments were performed remotely on CloudPendulum double-pendulum cell 203.

TABLE I
PLATFORM LIMITS VS. DEPLOYED SETTINGS.

Quantity	Platform limit	Deployed
Torque (measured)	0.15 Nm	main ≤ 0.13 ; 20 Nm/s slew
Passive-joint assist	forbidden	≤ 5 mNm (development only)
Velocity (measured)	50 rad/s	ref. ≤ 20 ; nominal peak 33.6
Joint position	$\pm 360^\circ$	within limit
Control rate	≤ 500 Hz	target 400; measured 358–375 Hz
Attempt length	60 s	10–18 s control windows

TABLE II
IDENTIFICATION DATA CAMPAIGNS (CELL 203, 200 Hz).

Trajectory set	Runs	Length (s)	Peak $ \dot{q} $ (rad/s)	Aborts
Free-swing, $q_1(0) \in [0.6, 2.4]$ rad	5	≤ 6.0	4.7–37.5	1
D-optimal multisine (A/B)	2	4.0	6.2–11.9	0
High-speed pilots (unguarded)	2	≤ 0.8	39.4–39.9	2
Burst chirps (velocity guard)	10	3.0	11.5–20.2	0

and upright target $x^* = [\pi, 0, 0, 0]^\top$. The *acrobot* drives the elbow, the *pendubot* the shoulder. Table I contrasts the platform limits with our deliberately conservative deployment; all experiments use the same physical cell (id 203).

III. SYSTEM IDENTIFICATION

A. Data Collection

All data were collected on cell 203 by a dedicated low-torque collector ($|\tau| \leq 60$ mNm, 200 Hz, per-step measured time stamps); Table II summarizes the four campaigns. Free swings constrain gravity, inertia, and passive friction; the D-optimal multisine (coefficients maximizing log det of the Fisher information, collected twice with different phases so one run is held out) constrains coupling and actuated-joint friction. The two unguarded high-speed pilots tripped the velocity abort within 0.8 s; the subsequent burst chirps add a soft velocity guard (torque fade above 22 rad/s) and reached 11.5–20.2 rad/s without a single abort, exciting the velocity-dependent friction terms needed for horizon-length validation. The merged set has 16 usable trajectories (~ 10.6 k samples).

B. Identified Model

Individual link parameters (m_i, l_i, r_i, I_i) are structurally inseparable from input–output data, so *base parameters* are identified instead. The model is

$$M(q)\ddot{q} + h(q, \dot{q}) + G(q) + \tau_f(\dot{q}) = \tau, \quad (1)$$

TABLE III

IDENTIFIED BASE PARAMETERS OF CELL 203. INERTIA TERMS $P_{1,2,3,6}$ ARE IN kg m^2 , GRAVITY TERMS $P_{4,5}$ AND c_f IN Nm , AND b IN Nm s/rad .

P_1	$6.50 \text{e-}4$	P_4	$9.07 \text{e-}2$	c_{f1}	$1.72 \text{e-}3$
P_2	$1.75 \text{e-}4$	P_5	$3.16 \text{e-}2$	c_{f2}	$3.91 \text{e-}3$
P_3	$1.84 \text{e-}4$	b_1	$2.81 \text{e-}4$	d_1	≈ 0
P_6	$2.06 \text{e-}4$	b_2	$1.04 \text{e-}6$	d_2	$7.9 \text{e-}7$

where $h = [-P_2 \sin q_2 (2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2), P_2 \sin q_2 \dot{q}_1^2]^\top$ and

$$\begin{aligned} M_{11} &= P_1 + 2P_2 \cos q_2, & M_{12} &= P_3 + P_2 \cos q_2, \\ M_{22} &= P_6, & G_1 &= P_4 \sin q_1 + P_5 \sin(q_1 + q_2), \\ G_2 &= P_5 \sin(q_1 + q_2), \end{aligned} \quad (2)$$

with P_6 independent of P_3 to absorb motor/rotor inertia on the elbow axis, and per-joint smooth friction

$$\tau_{f,i} = b_i \dot{q}_i + c_{f,i} \tanh(\dot{q}_i/\varepsilon) + d_i \dot{q}_i \sqrt{\dot{q}_i^2 + 0.25}, \quad \varepsilon = 0.05. \quad (3)$$

The 12 parameters are fitted by windowed simulation-error minimization (80-step RK4 windows re-anchored at measured states, measured per-step Δt), solved by IPOPT with a trust-region least-squares fallback [8], [9]. Table III lists the identified values. The fact that P_6 exceeds P_3 by 12% is consistent with additional inertia on the elbow axis, while the elbow Coulomb friction $c_{f2} = 3.9 \text{ mNm}$ is $2.3\times$ the shoulder value. A normalized Fisher-information analysis (condition number 2.2×10^2) flags c_{f1} and d_1 as weakly identifiable; the quadratic damping terms fit to ≈ 0 .

C. Validation Metrics

Two metrics tie model quality to control. *Single-step prediction*: during every hardware run, $(x[k], \tau[k])$ predicts $\hat{x}[k+1]$ over the measured loop period; the residual is logged (median/p95). *MPC-horizon prediction*: replaying measured inputs open-loop from many anchors gives a p95 angle error of approximately 12° at $18 \times 20 \text{ ms} = 0.36 \text{ s}$. This result motivated limiting the prediction horizon to 0.36 s ; it is not, by itself, a closed-loop stability guarantee. On held-out trajectories the open-loop cross-validation RMSE (q_1/q_2) is $8.0/10.9^\circ$ over the full 6 s free swing from $q_1(0) = 1.9 \text{ rad}$ and $4.3/6.6^\circ$ over the held-out high-speed chirp (Fig. 1).

IV. CONTROL METHOD

A. NMPC Formulation

A swing-up reference is generated offline by multiple-shooting OCP (IPOPT): minimize $\int (\tau_1^2 + \tau_2^2) dt$ over $T=3 \text{ s}$, $N=150$ RK4 steps ($\Delta t_{\text{ref}}=20 \text{ ms}$), subject to the identified dynamics, torque bounds, $|\dot{q}| \leq 20 \text{ rad/s}$, and an exact upright terminal state, followed by a 3 s hold. Online, an acados SQP-RTI solver [7] tracks it with a least-squares cost ($Q=\text{diag}(120, 50, 2, 2)$, $Q_f=\text{diag}(3000, 1200, 80, 80)$, $R\approx\text{diag}(0.05, 0.1)$), horizon $N_{\text{MPC}}=18$ (0.36 s), hard input and soft velocity bounds, and one QP iteration per cycle at a target period of 2.5 ms . Velocities are estimated by fusing the hardware signal with wrapped position differences (median

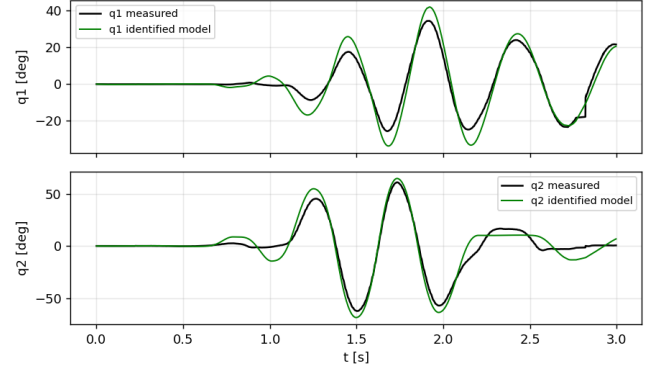


Fig. 1. Cross-validation on the held-out high-speed excitation `oid_hi_long_09`: measured angles (black) vs. open-loop simulation with the identified model (green).

and exponential moving-average filters). For the pendubot, a 20 Nm/s first-stage slew-rate constraint limits measured-torque overshoot during shoulder reversals.

B. Micro-Compensation Strategy

The passive joint receives a bounded assist during development: the acrobot's shoulder and the pendubot's elbow are limited to 5 mNm , i.e., 3.8% of the main bound. This is of the same order as the identified Coulomb friction and can therefore materially alter capture and upright behavior. It must not be interpreted as a negligible perturbation. *Passive-joint friction compensation is forbidden during evaluation; all assisted results below are development-only diagnostics.*

C. Disturbance Tests

Development-stage robustness was probed with the platform's server-side disturbance API: a timed *position* jump to $[\pi - 0.25, 0.25] \text{ rad}$ at $t=5 \text{ s}$ (both configurations), and, for the pendubot, *torque* pulses of $\pm 15 \text{ mNm}$ on the passive elbow at $t=7$ and 9 s . These scheduled tests are not equivalent to the randomized evaluation protocol.

V. EXPERIMENTAL RESULTS

A. Nominal Swing-Up and Stabilization

Table IV summarizes one recorded nominal run per configuration ($\sim 18 \text{ s}$, measured loop periods $2.67\text{--}2.69 \text{ ms}$). Both assisted controllers swing up and remain near upright without aborting in these runs. Whole-run single-step error is $\text{p95} \leq 0.35^\circ$, and solve p99 is $\leq 1.4 \text{ ms}$. Figs. 2 and 3 show the corresponding pendubot and acrobot traces. Approximate first arrivals near upright occur at 2 s for the acrobot and 4.5 s for the pendubot; these visually estimated times are not competition up-time metrics.

B. Disturbance Recovery

All three logged assisted disturbance runs ended near upright without aborting (Table V). The reported p95 prediction errors are computed over each complete run, not over disturbance-only windows. The acrobot position jump reached 46 rad/s filtered (42 rad/s raw), within the official

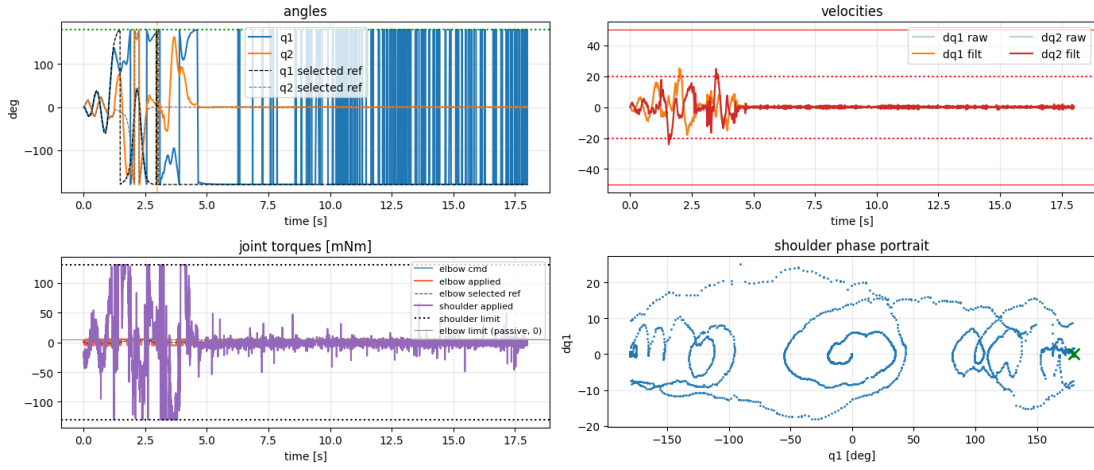


Fig. 2. Representative assisted pendubot run. Panels show wrapped joint angles and selected references, velocity estimates, commanded/applied torques, and the shoulder phase portrait. Angle-wrap discontinuities appear as vertical lines and are not physical jumps.

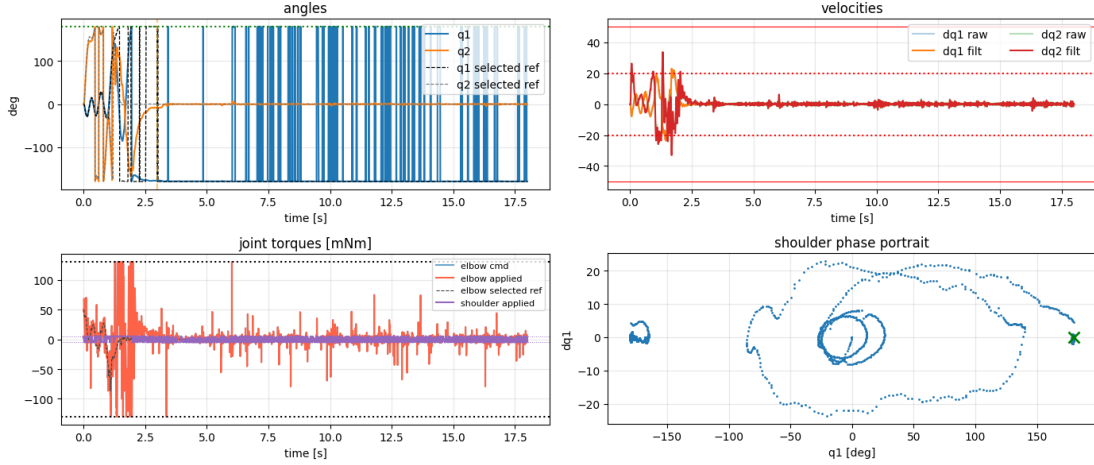


Fig. 3. Representative assisted acrobot run, with the same panels and conventions as Fig. 2. Angle-wrap discontinuities are plotting artifacts rather than physical jumps.

TABLE IV

SELECTED DEVELOPMENT-STAGE ASSISTED HARDWARE RUNS (CELL 203; ONE RUN PER CONFIGURATION).

	Acrobot	Pendubot
Loop steps / aborted	6732 / no	6682 / no
Final $ q_1 - \pi $ (deg)	0.26	0.13
Peak $ \dot{q} $ (rad/s)	33.6	25.1
Main torque (applied, mNm)	130 (elbow)	130 (shoulder)
Micro-assist (applied, mNm)	5 (shoulder)	5 (elbow)
Failed solves	2/6732	0/6682
1-step q err p95 (deg)	0.35	0.34
1-step $\dot{q}$ err p95 (rad/s)	0.65	0.53
Solve time mean / p99 (ms)	0.51 / 1.38	0.54 / 1.29
Mean loop period (ms)	2.67	2.69

TABLE V

DISTURBANCE-RECOVERY RUNS (ALL NON-ABORTING).

	Acrobot	Pendubot	
	pos. jump	torque pulses	pos. jump
Final $ q_1 - \pi $ (deg)	0.15	0.10	0.20
Peak $ \dot{q} $ (rad/s)	46.0	26.3	31.9
Peak main applied (mNm)	130.0	130.0	130.0
1-step q p95 (deg)	0.91	0.82	0.37
1-step $\dot{q}$ p95 (rad/s)	2.42	0.90	0.54
Failed solves	3/3590	0/6484	10/6637
Mean loop period (ms)	2.79	2.78	2.71

VI. DISCUSSION

One-step accuracy does not explain the failure alone.

50 rad/s velocity limit; it remains development-only because of the passive-joint assist, not because of a velocity violation. An acrobot *torque*-disturbance run exists only as video and is not included as quantitative evidence.

In four strictly uncompensated pendubot experiments (elbow torque identically zero), the model achieved a median single-step error of 0.065° before departure. The measured trajectory left the reference path at 0.30–0.35 s and exceeded 30° path distance at 0.866–0.875 s in the three runs long enough

to reach that threshold. No actuator saturation occurred before the initial departure. These observations reduce the likelihood of gross one-step model or torque-scale error, but do not exclude finite-horizon model error, state-estimation effects, or limited trackability.

Role of the passive-joint assist. The 5 mNm bound is comparable with the identified passive-joint Coulomb friction. Assisted runs succeeded where the tested uncompensated pendubot runs did not, so friction is a plausible contributor. This comparison is diagnostic rather than a controlled ablation: the available runs do not isolate the assist from all reference and implementation changes.

Reference design is a remaining hypothesis. Capping the uncompensated reference OCP at 80 mNm produced a 200 ms torque flat-top at $t=0.34\text{--}0.52$ s, overlapping the observed departure window. Offline re-optimization with a 130 mNm cap removes the flat-top while changing $\int \tau^2 dt$ from 0.00439 to 0.00434. This correlation motivates a hardware ablation, but does not yet establish the flat-top as the cause of failure.

VII. LIMITATIONS AND FUTURE WORK

(i) Passive-joint compensation is prohibited during evaluation, so the assisted runs do not establish qualification performance; strictly single-actuator control remains open. (ii) The experiments do not provide the required randomized-disturbance up-time, baseline comparison, or automatic commissioning time on an unseen cell. (iii) Recovery time cannot be reconstructed from the retained aggregate outputs, and the acrobot torque-disturbance run lacks a numerical log. Future work should prioritize an evaluation-compliant hardware run, event-aligned logging, automatic adaptation to unseen cells, and reference designs that avoid long constraint-active segments.

VIII. CONCLUSION

A base-parameter model of CloudPendulum cell 203 was identified from hardware data and validated by

held-out open-loop rollouts and in-run single-step errors. Development-stage NMPC runs with a bounded 5 mNm passive-joint assist reached and held the upright neighborhood in both configurations and ended near upright after scheduled disturbances. Because the assist is forbidden in evaluation, these results diagnose the influence of passive-joint friction but do not constitute qualification performance. The retained uncompensated runs show a reproducible departure from the tested reference; separating finite-horizon model error, friction, and reference trackability requires a compliant hardware ablation with event-aligned metrics.

REFERENCES

- [1] S. Kumar, “Swinging pendulums on the cloud: Digitalization of simulation and experimental infrastructure for feedback-based active learning,” in *Chalmers Conference on Teaching and Learning (KUL)*, 2025.
- [2] F. Wiebe, S. Kumar, L. J. Shala, S. Vyas, M. Javadi, and F. Kirchner, “Open source dual-purpose acrobot and pendubot platform: Benchmarking control algorithms for underactuated robotics,” *IEEE Robot. Autom. Mag.*, vol. 31, no. 2, pp. 113–124, Jun. 2024, doi: 10.1109/MRA.2023.3341257.
- [3] F. Wiebe, S. Vyas, L. J. Maywald, S. Kumar, and F. Kirchner, “RealAIGym: Education and research platform for studying athletic intelligence,” in *Proc. Robot. Sci. Syst. Workshop Mind the Gap*, New York, NY, USA, 2022.
- [4] F. Wiebe *et al.*, “Reinforcement learning for athletic intelligence: Lessons from the 1st AI Olympics with RealAIGym competition,” in *Proc. 33rd Int. Joint Conf. Artif. Intell.*, 2024, art. no. 1043, pp. 8833–8837, doi: 10.24963/ijcai.2024/1043.
- [5] M. W. Spong, “The swing up control problem for the acrobot,” *IEEE Control Systems Magazine*, vol. 15, no. 1, pp. 49–55, 1995.
- [6] M. W. Spong and D. J. Block, “The Pendubot: A mechatronic system for control research and education,” in *Proc. 34th IEEE Conf. Decision and Control*, 1995, pp. 555–556.
- [7] R. Verschuere, G. Frison, D. Kouzoupis, J. Frey, N. van Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, “acados—a modular open-source framework for fast embedded optimal control,” *Mathematical Programming Computation*, vol. 14, pp. 147–183, 2022.
- [8] A. Wächter and L. T. Biegler, “On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming,” *Mathematical Programming*, vol. 106, no. 1, pp. 25–57, 2006.
- [9] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, “CasADi: A software framework for nonlinear optimization and optimal control,” *Mathematical Programming Computation*, vol. 11, no. 1, pp. 1–36, 2019.